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Construction Enterprise Data Platform and Big Data Analytics for Efficient Workforce Management

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***Abstract*—To enhance workforce management in the construction industry, we integrated enterprise data platforms with big data analytics. The improved platform addresses issues such as inefficiencies in traditional workforce management practices, data silos, and safety concerns. Big data analytics optimized labor allocation and safety protocols, and reduced operational costs. To provide advanced analytics tools, real-time data monitoring, and comprehensive employee training programs on the platform, we examined successful case studies. The feasibility of the improved platform was validated by the implementation cases, which demonstrated significant improvements in efficiency and safety enhancement. The results of this study also prove the importance of the data-driven approach in workforce management of the construction industry.**

Keywords*—big data, workforce management, construction industry, data analytics

I. Introduction

Effective workforce management is critical in the construction industry, where personnel dedicate significant working hours to vast and intricate projects [1]. With the increasing cost of hiring skilled workers, construction organizations face challenges in managing the workforce, controlling risks, and adhering to project schedules. Conventional approaches to workforce management rely on paper-based systems and discrete data, frequently leading to inefficiencies and increased expenses [2]. Consequently, an innovative strategy is required. In this study, we integrated big data analytics in a construction enterprise data platform following a transformative trend to enable effective management of the immense volume of data generated throughout a project's lifecycle.

Construction enterprise data includes employee records, project schedules, and equipment performance [3]. By applying big data analytics, construction enterprise can make informed decisions regarding workforce management. In this study, the significance of capturing, analyzing, and utilizing construction enterprise data platforms and big data analytics in optimizing construction workforce management was also discussed.

II. Literature Review

A. Big Data Analytics

Big data analytics are used to improve construction processes and projects and enhance management and performance (Fig. 1). Through a systematic analysis, decision-making effectiveness can be enhanced through the real-time evaluation of project performance and resource allocation. When effectively applied, big data analytics is used to address significant workforce management challenges, such as labor scheduling and conflicts with safety requirements, by leveraging the vast, disparate data volumes generated in the construction industry. Big data analytics is expected to reduce costs in engineering and construction by 21% over the next decade, demonstrating its financial benefits [5]. Despite these advantages, the adoption of big data analytics in construction lags behind other industries, suggesting a need for increased research and practical applications.

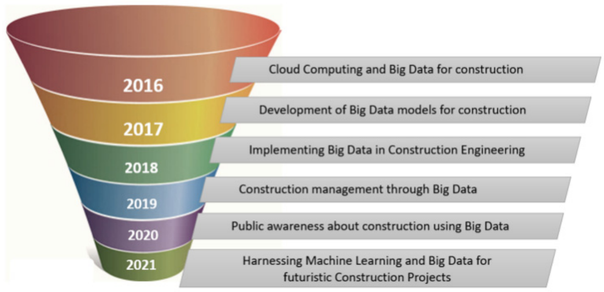

Fig. 1. Funnel diagram showing trends in big data research in construction since 2016 [4].

The construction industry faces challenges in human resource management (HRM), grappling with numerous pressures related to workforce administration [6]. Traditional project management approaches suffer from fragmented information systems, leading to inefficiencies. Big data analytics offers solutions to these challenges by predicting and tracking labor resources. For example, the integration of Internet of Things (IoT) devices and building information modeling (BIM) enhances the tracking ability of worker productivity and site conditions, enabling proactive adjustments to project delivery (Fig. 2) [7].

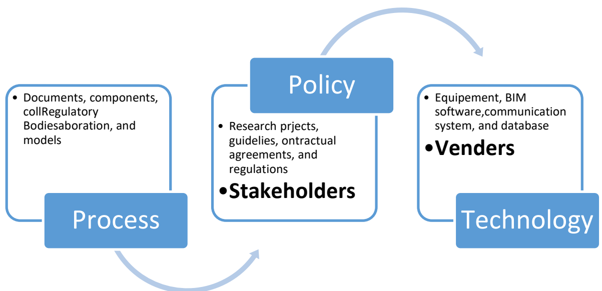

Fig. 2. Three main fields in BIM [8].

B. Effectiveness of Data-Driven Decision-Making

Previous studies verified the impact of big data analytics on performance [9,10]. Construction enterprises that integrate big data into their decision-making processes reported positive effects on resource management, cost reduction, and improved project duration (Fig. 3) [11]. Enterprises leveraging predictive analytics are better equipped to forecast labor demands and potential labor shortages. This is a significant advantage, considering unstable demand for qualified personnel and short-term project contracts. Moreover, analyzing historical performance data assists construction managers in evaluating trends, thereby enhancing overall project efficiency from initial construction through to handover.

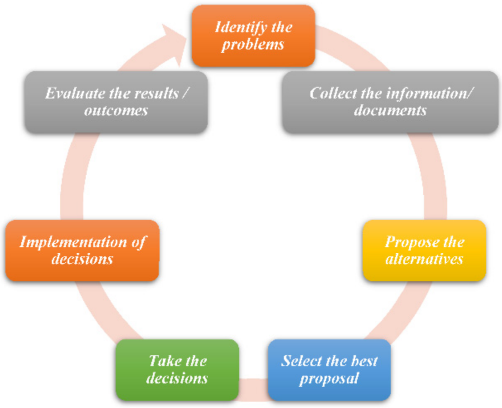

Fig. 3. General decision-making process.

III. Theoretical Framework

A. Current Theories

Bi-directional amplifiers (BDAs) in construction have been developed to provide foundational knowledge regarding big data utilization to enhance construction project management and operations. Theoretical structural models, such as the technology acceptance model (TAM), are used to examine users' perceptions and adoption of technologies in the workplace (Fig. 4) [12]. In the construction industry, TAM helps understand the factors influencing project managers' and workers' adoption of big data tools. The resource-based view (RBV) theory posits that organizations can gain a competitive advantage by employing valuable resources, including data analytics. Therefore, it is necessary to adopt robust data systems and facilitate efficient workforce planning and informed decision-making becomes clear.

The dynamic capability framework (Fig. 5) is used to evaluate an organization's capacity to operate and sustain innovation effectively in dynamic environments. The framework elucidates how enterprises can leverage big data analytics to address market fluctuations, labor shortages, and project-specific challenges.

The previous methods are used to construct mechanisms using big data analytics for transformative changes in construction workforce management.

Fig. 4. Big data innovation and diffusion in projects teams.

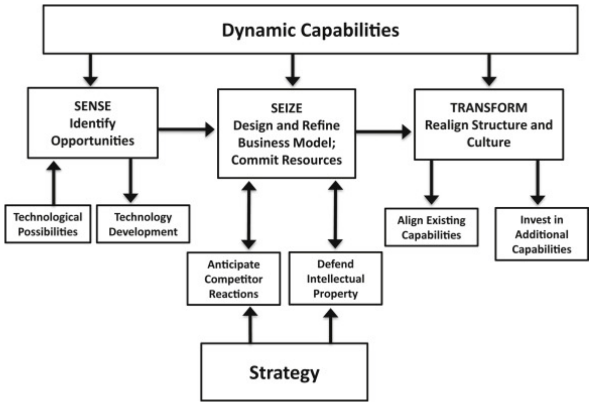

Fig. 5. Business models and dynamic capabilities.

B. New Method Development

We developed a new framework with big data analytics in workforce management of the construction industry. The big data-driven workforce optimization model (BDWOM) is used for data gathering, analysis, and decision-making on workforce management. The model uses big data to enhance labor forecasts, employee schedules, and resource management. Overall, BDWOM with a structured methodology is used to leverage big data. The second framework used in this study is the data-driven safety management framework (DDSMF) to improve safety through risk prediction and detection. This framework utilizes big data from various sources, including IoT devices and wearable technologies, to analyze potential hazards and enhance workers' safety. Consequently, construction enterprises can cultivate a robust safety culture through effective data analysis, thereby reducing accident risks for the workforce and enhancing employee well-being..

IV. Technical Framework

A. Data Platforms

In construction, several data platforms are used to process large amounts of information during project life cycles. One such platform is BIM (Fig. 6) to develop accurate models with physical structures. BIM provides an environment for stakeholders' cooperation and a source of information regarding design, materials, and the estimated amount of labor. Another platform is the enterprise resource planning (ERP) system that unite many business activities such as project management, procurement, and human resources. The ERP system helps construction enterprises achieve efficiency and productivity while enhancing department data flow.

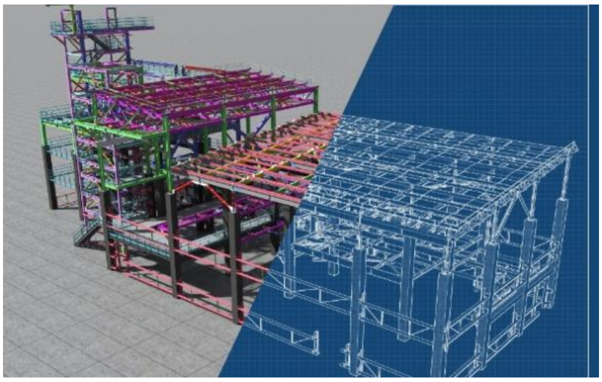

Fig. 6. BIM application.

Cloud-based project management solutions have recently gained significant traction within the industry due to their ability to make project data accessible from any location [13].

These platforms facilitate the mobilization of field employees, providing organizations with real-time information on workforce status and resource utilization. Collectively, these platforms are used for modern construction management, empowering enterprises to leverage data effectively for enhanced decision-making.

B. Role of Data Analytics

Data analytics plays a pivotal role in refining decision-making frameworks within the construction industry's workforce management [14]. To derive actionable information, construction managers employ various analytical methods for data gathering from diverse platforms and channels. On the platforms, valuable information is obtained for planning and adapting the construction management process [15]. For instance, regarding labor demand, predictive analytics is used to estimate future requirements and minimize overhead costs associated with overstaffing or delayed project delivery.

V. Comparison between Traditional and Proposed Approaches

A. Traditional Workforce Management

Traditional practices for workforce management in the construction industry rely on paper-based systems, disparate stand-alone tools, and informal, subjective processes. These methods involve spreadsheet utilization for scheduling and tracking labor hours or resources, which is inefficient and prone to errors. Communication is conducted through face-to-face meetings or phone calls, leading to time consumption and delayed information flow. Moreover, conventional techniques cannot monitor real-time project data, hindering prompt responses to dynamic changes in project conditions or labor demand [16]. Such disadvantages result in overstaffing or understaffing, elevated labor costs, and increased risks on construction sites. They also impede efficiency and adversely impact project timelines and costs.

B. Proposed Big Data-Driven Approaches

Big data, as presented in current and emerging proposals, offers a strategic and effective method for workforce management. By leveraging integrated systems and advanced technology, construction enterprises can obtain real-time information concerning labor productivity, project progress, and resource utilization [17]. Such solutions facilitate the accurate prediction of labor requirements and adjustments to staffing levels to meet project needs. Moreover, big data platforms enhance communication among team members by providing centralized access to all necessary information.

Furthermore, big data analytics fosters an understanding of workforce trends and patterns, empowering managers to implement actions that improve productivity and safety [18]. For example, real-time monitoring of operations helps identify specific delays or potential safety hazards before they escalate. In contrast to past practices characterized by limited visibility, inaccurate tracking, and slow decision-making, big data analytics enable construction enterprises to optimize their operations and enhance overall project outcomes. It also improves human capital management and promotes positive transformation within the sector's workforce.

VI. Case Studies and Analysis

A. Big Data in Workforce Management

Examples of how big data improved construction workforce management were studied. An example of operational success was Enterprise XYZ, which applied big data analytics to enhance safety on construction sites. The enterprise collects data on real-time safety risks through wearables, IoT sensors, and surveillance cameras. The analytics platform with machine learning algorithms recognizes patterns correlated with unsafe practices, so competent supervisors intervene. The platform has reduced workplace accidents by 30% in the first year of the platform's introduction [19]. In this case (Table I), real-time data analysis in safety risk identification enhanced the safety condition of workers and provided accountability. These changes toward a culture of safety involved big data analytics to minimize incident occurrences while improving other aspects of a project [20].

TABLE I. Case Study

Case Study	Key Focus Area	Outcome	Reference
Enterprise XYZ	Safety Management	30% reduction in workplace accidents	Neuroject, 2024
Enterprise ABC	Material Management	25% decrease in material costs	Neuroject, 2024

B. Data Analysis

Table II presents data on workforce analytics in construction organizations to examine the practical and conceptual understanding of workforce planning. The results are categorized as "Yes" or "No" with frequencies and percentages. Many organizations possessed ill-defined workforce planning strategies and inadequately prioritized workforce analytics. 25% of the organizations integrated multiple business process data sources for workforce scheduling decisions, indicating a lack of comprehensive data-driven workforce management [23].

TABLE II. Workforce Data Analytics On Production Efficiency [23]

Workforce Analytics	Yes		No	
	Frequency	%	Frequency	%
Does your organization have a well understood workforce planning framework?	6	30	14	70

Is the production department a major focus when it comes to workforce planning and analytics?	6	30	14	70
Is it necessary to combine more than one business process data when taking workforce scheduling decisions?	5	25	15	75

C. Comparative Research

Conventional workforce management methods, characterized by paperwork, manual processes, and fragmented information, result in inefficiencies, inaccuracies, and time waste. In contrast, big data-driven methods enhance accuracy, speed, and decision-making through real-time analytics. For instance, traditional methods hardly identify early safety risks or overstaffing due to a lack of integrated monitoring. However, big data systems leverage IoT devices, artificial intelligence, and predictive tools to optimize labor allocation, making projects safer, more cost-effective, and faster. Big data analytics helps reduce workforce costs by a third within a decade, demonstrating its superior effectiveness over conventional methods.

VII. Challenges and Recommendations

A. Challenges

1) Technical and Operational Challenges

In this study, we identified technical and operational challenges associated with implementing big data technology in the construction industry. A primary challenge is data aggregation from diverse sources, often originating from legacy systems not designed for advanced analytics. This lack of interoperability results in data silos, wherein significant information remains isolated within specific systems, limiting access for comprehensive analysis.

2) Data Privacy and Security Concerns

Data privacy and protection issues are paramount when utilizing big data tools for workforce management. The construction industry processes sensitive employee information, such as identification numbers and health records, necessitating robust protection against unauthorized third-party access and cyberattacks [21]. Compliance with legal frameworks, such as the General Data Protection Regulation (GDPR), which mandate standards for the collection, storage, and processing of personal information, is essential.

B. Future Recommendations

1) Integrating Big Data in Workforce Management

A phased approach needs to be adopted, beginning with an evaluation of the enterprise's capacity for big data integration and identifying areas requiring improvement. It is necessary to deploy user-friendly analytics and interfaces for easy data consolidation from various sources, including project management tools and IoT devices. Additionally, organizations need to implement training programs to equip employees with the skills to analyze and utilize data for informed decision-making.

2) Data Collection and Analysis

Enhancing construction workforce management necessitates enhanced data collection and analysis strategies [22]. For instance, leveraging mobile technology applications enables inputting real-time data on labor activities and site conditions, significantly improving the accuracy of recorded details. Furthermore, emerging technologies, such as machine learning, need to be used to uncover unrecognized patterns in data to enable optimal staffing and enhance safety protocols.

VIII. Conclusion

Data platforms and big data analytics have significant potential to enhance workforce management in the construction industry. By leveraging advanced analytics, enterprises can overcome the limitations of conventional methods, which are constrained by inefficiencies and information silos. Therefore, a new platform was developed to improve labor utilization and safety measures and reduce expenditures. Big data solutions yielded substantial returns, increasing safety and reducing accidents and material losses. However, it is required to address challenges such as data integration, privacy concerns, and the scarcity of skilled human resources. Big data analytics in workforce management must be integrated with innovative data collection approaches. As the construction industry evolves, the strategic adoption of big data analytics has become increasingly crucial for operation enhancement.

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